Chapter 11. The Analysis of Covariance (ANACOVA)

Contents

1 Introduction

- ANACOVA combines the features of ANOVA and regression.
- It reduces a large error variance in the ANOVA model by adding one or more quantitative variables (covariates) linearly related to the response.
- Setting: besides the response y there is a covariate x, linearly related to y, that *cannot be controlled* by the experimenter but can be *observed* along with y (recall Ch. 4: known but uncontrollable nuisance factor).

Example 11.1: Cracker promotion

A company studies three promotion types on cracker sales. 15 stores; each store is randomly assigned to one promotion (5 stores per type). y = number of cases sold during the promotional period; x = sales in the preceding period (covariate).

	Store 1		2		3		4		5	
Trt	y	x	y	x	y	x	y	x	y	x
1	38	21	39	26	36	22	45	28	33	19
2	43	34	38	26	38	29	27	18	34	25
3	24	23	32	29	31	30	21	16	28	29

The scatter of y around the treatment averages is large, but the scatter around the *treatment regression lines* (on x) is much smaller: part of the "error" is explained by x.

2 Model

Definition 11.2

$$y_{ij} = \mu + \tau_i + \beta(x_{ij} - \bar{x}_{\cdot\cdot}) + \varepsilon_{ij}, \qquad i = 1, \ldots, a;\ j = 1, \ldots, n,$$

x_{ij} given (fixed), $\sum_i \tau_i = 0$, $\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$, β = common regression coefficient (slope). Hence

$$\mathrm{E}(y_{ij}) = \mu + \tau_i + \beta(x_{ij} - \bar{x}_{\cdot\cdot}), \qquad \mathrm{Var}(y_{ij}) = \sigma^2.$$

Each treatment has its own regression line, all **parallel** with slope β; τ_i is the vertical shift.

2.1 Notation: corrected sums of squares and cross products

$$S_{xx} = \sum_i \sum_j (x_{ij} - \bar{x}_{\cdot\cdot})^2, \qquad S_{xy} = \sum_i \sum_j (x_{ij} - \bar{x}_{\cdot\cdot})(y_{ij} - \bar{y}_{\cdot\cdot}), \qquad S_{yy} = \sum_i \sum_j (y_{ij} - \bar{y}_{\cdot\cdot})^2,$$

$$T_{xx} = n \sum_i (\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot})^2, \qquad T_{xy} = n \sum_i (\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot})(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}), \qquad T_{yy} = n \sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2,$$

$$E_{xx}=\sum_i\sum_j(x_{ij}-\bar{x}_{i\cdot})^2,\qquad E_{xy}=\sum_i\sum_j(x_{ij}-\bar{x}_{i\cdot})(y_{ij}-\bar{y}_{i\cdot}),\qquad E_{yy}=\sum_i\sum_j(y_{ij}-\bar{y}_{i\cdot})^2,$$

with $S=T+E$ in each case (same proof as the one-way ANOVA identity, applied to products).

3 Least Squares Estimation

Derivation

$S=\sum_i\sum_j\left(y_{ij}-\mu-\tau_i-\beta(x_{ij}-\bar{x}_{\cdot\cdot})\right)^2$.

$$\frac{\partial S}{\partial\mu}=0\Rightarrow\sum_i\sum_j\left(y_{ij}-\hat{\mu}-\hat{\tau}_i-\hat{\beta}(x_{ij}-\bar{x}_{\cdot\cdot})\right)=0\Rightarrow\hat{\mu}=\bar{y}_{\cdot\cdot},$$

$$\frac{\partial S}{\partial\tau_i}=0\Rightarrow n\hat{\mu}+n\hat{\tau}_i+\hat{\beta}\sum_{j=1}^{n}(x_{ij}-\bar{x}_{\cdot\cdot})=y_{i\cdot}\Rightarrow\hat{\tau}_i=(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot})-\hat{\beta}(\bar{x}_{i\cdot}-\bar{x}_{\cdot\cdot}),$$

$$\frac{\partial S}{\partial\beta}=0\Rightarrow\sum_i\sum_j\left(y_{ij}-\hat{\mu}-\hat{\tau}_i-\hat{\beta}(x_{ij}-\bar{x}_{\cdot\cdot})\right)(x_{ij}-\bar{x}_{\cdot\cdot})=0.$$

Theorem 11.3

$$\hat{\beta}=\frac{E_{xy}}{E_{xx}},\qquad\hat{\mu}=\bar{y}_{\cdot\cdot},\qquad\hat{\tau}_i=(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot})-\hat{\beta}(\bar{x}_{i\cdot}-\bar{x}_{\cdot\cdot}).$$

Proof. (In $\partial S/\partial\mu$, $\sum\hat{\tau}_i=0$ and $\sum\sum(x_{ij}-\bar{x}_{\cdot\cdot})=0$ give $N\hat{\mu}=y_{\cdot\cdot}$.) Expand the β-equation:

$$\sum\sum y_{ij}(x_{ij}-\bar{x}_{\cdot\cdot})-\hat{\mu}\underbrace{\sum\sum(x_{ij}-\bar{x}_{\cdot\cdot})}_{=0}-\sum_i\hat{\tau}_i\sum_j(x_{ij}-\bar{x}_{\cdot\cdot})-\hat{\beta}\sum\sum(x_{ij}-\bar{x}_{\cdot\cdot})^2=0.$$

Since $\sum\sum y_{ij}(x_{ij}-\bar{x}_{\cdot\cdot})=S_{xy}$ and $\sum_j(x_{ij}-\bar{x}_{\cdot\cdot})=n(\bar{x}_{i\cdot}-\bar{x}_{\cdot\cdot})$,

$$\sum_i\hat{\tau}_i\,n(\bar{x}_{i\cdot}-\bar{x}_{\cdot\cdot})+\hat{\beta}S_{xx}=S_{xy}.$$

Substitute $\hat{\tau}_i$:

$$\underbrace{n\sum_i(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot})(\bar{x}_{i\cdot}-\bar{x}_{\cdot\cdot})}_{=T_{xy}}-\hat{\beta}\underbrace{n\sum_i(\bar{x}_{i\cdot}-\bar{x}_{\cdot\cdot})^2}_{=T_{xx}}+\hat{\beta}S_{xx}=S_{xy}$$

$$\Rightarrow\hat{\beta}(S_{xx}-T_{xx})=S_{xy}-T_{xy}\;\Rightarrow\;\hat{\beta}=\frac{E_{xy}}{E_{xx}}.$$

So the slope is estimated *within* treatments (pooled within-group regression). ■

Theorem 11.4: Error sum of squares

$$SS_E=E_{yy}-\frac{E_{xy}^2}{E_{xx}},\qquad\text{df}=a(n-1)-1,\qquad\text{Var}(\hat{\beta})=\frac{\sigma^2}{E_{xx}}.$$

Proof. Fitted value $\hat{y}_{ij}=\hat{\mu}+\hat{\tau}_i+\hat{\beta}(x_{ij}-\bar{x}_{\cdot\cdot})=\bar{y}_{i\cdot}+\hat{\beta}(x_{ij}-\bar{x}_{i\cdot})$, so $e_{ij}=(y_{ij}-\bar{y}_{i\cdot})-\hat{\beta}(x_{ij}-\bar{x}_{i\cdot})$ and

$$SS_E=\sum\sum e_{ij}^2=E_{yy}-2\hat{\beta}E_{xy}+\hat{\beta}^2E_{xx}=E_{yy}-2\frac{E_{xy}^2}{E_{xx}}+\frac{E_{xy}^2}{E_{xx}}=E_{yy}-\frac{E_{xy}^2}{E_{xx}}.$$

Parameters: μ, $a-1$ free τ's, $\beta \Rightarrow a+1$; df $= an-a-1 = a(n-1)-1$. Variance: $\hat\beta = \sum\sum(x_{ij}-\bar x_{i\cdot})y_{ij}/E_{xx}$ is linear in independent y's, so $\mathrm{Var}(\hat\beta) = \sigma^2\sum\sum(x_{ij}-\bar x_{i\cdot})^2/E_{xx}^2 = \sigma^2/E_{xx}$. ■

Adjusted treatment means: $\hat\mu + \hat\tau_i = \bar y_{i\cdot} - \hat\beta(\bar x_{i\cdot} - \bar x_{\cdot\cdot})$ — the treatment mean adjusted to the common covariate value $\bar x_{\cdot\cdot}$.

4 Regression (Matrix) Formulation

Derivation

With $a=3$, $n=5$ and $\tau_3 = -\tau_1-\tau_2$, index the 15 observations by ℓ:

$$y_\ell = \mu + \tau_1 z_{1\ell} + \tau_2 z_{2\ell} + \beta(x_\ell - \bar x) + \varepsilon_\ell, \qquad z_1 = \begin{cases} 1 & \text{trt 1} \\ -1 & \text{trt 3} \\ 0 & \text{otherwise} \end{cases}, \; z_2 = \begin{cases} 1 & \text{trt 2} \\ -1 & \text{trt 3} \\ 0 & \text{otherwise} \end{cases},$$

$$\mathbf{X} = \left[\begin{array}{ccc|c} 1 & 1 & 0 & x_{11}-\bar x_{\cdot\cdot} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & 0 & x_{15}-\bar x_{\cdot\cdot} \\ \hline 1 & 0 & 1 & x_{21}-\bar x_{\cdot\cdot} \\ \vdots & & & \vdots \\ 1 & 0 & 1 & x_{25}-\bar x_{\cdot\cdot} \\ \hline 1 & -1 & -1 & x_{31}-\bar x_{\cdot\cdot} \\ \vdots & & & \vdots \\ 1 & -1 & -1 & x_{35}-\bar x_{\cdot\cdot} \end{array}\right], \qquad \hat{\boldsymbol\beta} = \begin{pmatrix} \hat\mu \\ \hat\tau_1 \\ \hat\tau_2 \\ \hat\beta \end{pmatrix} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y},$$

$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol\beta} = \mathbf{H}\mathbf{y}$, $\mathbf{e} = (\mathbf{I}-\mathbf{H})\mathbf{y}$, $SS_E = \mathbf{e}^T\mathbf{e} = \mathbf{y}^T(\mathbf{I}-\mathbf{H})\mathbf{y}$.

Source	DF	
Model $(\tau_1,\dots,\tau_{a-1},\beta)$	a	$F_0 = MS_{\text{Model}}/MS_E$
Error	$an-a-1$	
Total	$an-1$	

Cracker example (`lm`): $\hat\mu = 33.800$, $\hat\tau_1 = 6.017$, $\hat\tau_2 = 0.942$, $\hat\tau_3 = -6.959$, $\hat\beta = 0.899$.

5 Hypothesis Testing via Extra Sums of Squares

Testing of interest: (i) $H_0: \tau_1 = \cdots = \tau_a = 0$; (ii) $H_0: \beta = 0$.

Derivation

Full: $\mathbf{y} = \mathbf{X}\boldsymbol\beta + \boldsymbol\varepsilon = \mathbf{X}_1\boldsymbol\beta_1 + \mathbf{X}_2\boldsymbol\beta_2 + \boldsymbol\varepsilon$; Reduced: $\mathbf{y} = \mathbf{X}_2\boldsymbol\beta_2 + \boldsymbol\varepsilon$; $H_0: \boldsymbol\beta_1 = \mathbf{0}$:

$$F_0 = \frac{[SS_E(\mathrm{R}) - SS_E(\mathrm{F})]/[df_R - df_F]}{SS_E(\mathrm{F})/df_F} \overset{H_0}{\sim} F_{df_R-df_F,\,df_F}.$$

(i) Treatments. Full $y_{ij} = \mu + \tau_i + \beta(x_{ij}-\bar x_{\cdot\cdot}) + \varepsilon_{ij}$ ($df_F = an-a-1$); reduced $y_{ij} = \mu + \beta(x_{ij}-$

$\bar{x}_{\cdot\cdot}) + \varepsilon_{ij}$ (simple regression, $df_R = an - 2$), whose error SS is $S_{yy} - S_{xy}^2/S_{xx}$:

$$F_0 = \frac{\left[(S_{yy} - \frac{S_{xy}^2}{S_{xx}}) - (E_{yy} - \frac{E_{xy}^2}{E_{xx}})\right]/(a-1)}{SS_E(\mathrm{F})/[a(n-1)-1]} \overset{H_0}{\sim} F_{a-1,\,a(n-1)-1}.$$

(ii) Slope. Reduced $y_{ij} = \mu + \tau_i + \varepsilon_{ij}$ (one-way ANOVA, error SS E_{yy}, df $a(n-1)$):

$$F_0 = \frac{[E_{yy} - SS_E(\mathrm{F})]/1}{SS_E(\mathrm{F})/[a(n-1)-1]} = \frac{E_{xy}^2/E_{xx}}{MS_E} \overset{H_0}{\sim} F_{1,\,a(n-1)-1} \qquad \Big(= t_0^2,\ t_0 = \frac{\hat{\beta}}{\sqrt{MS_E/E_{xx}}} \Big).$$

Example 11.5: Cracker promotion, line by line

Totals: $x_{i\cdot} = 116, 132, 127$ ($x_{\cdot\cdot} = 375$, $\bar{x}_{\cdot\cdot} = 25$); $y_{i\cdot} = 191, 180, 136$ ($y_{\cdot\cdot} = 507$, $\bar{y}_{\cdot\cdot} = 33.8$); $\sum x^2 = 9735$, $\sum y^2 = 17783$, $\sum xy = 12937$.

$$\begin{aligned}
S_{xx} &= 9735 - \frac{375^2}{15} = 9735 - 9375 = 360.0,\\
S_{yy} &= 17783 - \frac{507^2}{15} = 17783 - 17136.6 = 646.4,\\
S_{xy} &= 12937 - \frac{375 \cdot 507}{15} = 12937 - 12675 = 262.0,\\
T_{xx} &= \frac{116^2 + 132^2 + 127^2}{5} - 9375 = 26.8,\\
T_{yy} &= \frac{191^2 + 180^2 + 136^2}{5} - 17136.6 = 338.8,\\
T_{xy} &= \frac{116(191) + 132(180) + 127(136)}{5} - 12675 = -37.4,\\
E_{xx} &= 360.0 - 26.8 = 333.2, \quad E_{yy} = 646.4 - 338.8 = 307.6, \quad E_{xy} = 262.0 + 37.4 = 299.4.
\end{aligned}$$

$$\begin{aligned}
\hat{\beta} &= \frac{E_{xy}}{E_{xx}} = \frac{299.4}{333.2} = 0.8986,\\
SS_E(\mathrm{F}) &= 307.6 - \frac{299.4^2}{333.2} = 307.6 - 269.03 = 38.57 \quad (11\text{ df}), \qquad MS_E = 3.506,\\
SS_E(\mathrm{R}_\tau) &= 646.4 - \frac{262^2}{360} = 646.4 - 190.68 = 455.72 \quad (13\text{ df}).
\end{aligned}$$

Treatments: $F_0 = \dfrac{(455.72 - 38.57)/2}{38.57/11} = \dfrac{208.58}{3.506} = 59.5 > F_{0.05,2,11} = 3.98$: reject; promotions differ (R: 59.42).

Slope: reduced model error SS $= E_{yy} = 307.6$ (12 df); $F_0 = \dfrac{(307.6 - 38.57)/1}{3.506} = 76.7 > F_{0.05,1,11} = 4.84$: reject $H_0 : \beta = 0$; the covariate matters. $se(\hat{\beta}) = \sqrt{3.506/333.2} = 0.103$.
Estimates: $\bar{x}_{i\cdot} = 23.2,\ 26.4,\ 25.4$; $\bar{y}_{i\cdot} = 38.2,\ 36.0,\ 27.2$:

$$\begin{aligned}
\hat{\tau}_1 &= (38.2 - 33.8) - 0.8986(23.2 - 25) = 4.4 + 1.617 = 6.017,\\
\hat{\tau}_2 &= (36.0 - 33.8) - 0.8986(26.4 - 25) = 2.2 - 1.258 = 0.942,\\
\hat{\tau}_3 &= (27.2 - 33.8) - 0.8986(25.4 - 25) = -6.6 - 0.359 = -6.959.
\end{aligned}$$

Adjusted means: 39.82, 34.74, 26.84. Compare with the one-way ANOVA ignoring x: $MS_E = 307.6/12 = 25.63$ vs 3.51 with the covariate — the error variance is reduced by a factor of 7.

6 Testing for Parallel Slopes

Derivation (Final 2025)

Model with treatment-specific slopes:

$$y_{ij} = \mu + \tau_i + \beta_i(x_{ij} - \bar{x}_{\cdot\cdot}) + \varepsilon_{ij}$$

$$\Rightarrow \; y_\ell = \mu + \tau_1 z_{1\ell} + \tau_2 z_{2\ell} + \beta_1(x_{1\ell} - \bar{x}) + \beta_2(x_{2\ell} - \bar{x}) + \beta_3(x_{3\ell} - \bar{x}) + \varepsilon_\ell,$$

where $x_{i\ell} - \bar{x}$ is the centered covariate if observation ℓ is in treatment i and 0 otherwise:

$$\mathbf{X} = \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & x_{1j} - \bar{x} & 0 & 0 \\ \hline 1 & 0 & 1 & 0 & x_{2j} - \bar{x} & 0 \\ \hline 1 & -1 & -1 & 0 & 0 & x_{3j} - \bar{x} \end{array}\right],$$

each row block having n rows, and

$$\boldsymbol{\beta} = (\mu, \tau_1, \tau_2, \beta_1, \beta_2, \beta_3)^T.$$

$H_0 : \beta_1 = \beta_2 = \cdots = \beta_a$ (parallel lines). Full model (separate slopes): each treatment is an ordinary simple regression, so

$$SS_E(\mathrm{F}) = \sum_{i=1}^{a} \left[E_{yy}^{(i)} - \frac{(E_{xy}^{(i)})^2}{E_{xx}^{(i)}}\right], \quad df_F = a(n-2),$$

with $E^{(i)}$ the within-treatment-i sums. Reduced model = common slope ANACOVA: $SS_E(\mathrm{R}) = E_{yy} - E_{xy}^2/E_{xx}$, $df_R = a(n-1) - 1$. Then

$$F_0 = \frac{[SS_E(\mathrm{R}) - SS_E(\mathrm{F})]/(a-1)}{SS_E(\mathrm{F})/[a(n-2)]} \overset{H_0}{\sim} F_{a-1,\, a(n-2)}.$$

Example 11.6: Parallelism for the cracker data

Within-treatment sums: trt 1: $E_{xx}^{(1)} = 54.8$, $E_{yy}^{(1)} = 78.8$, $E_{xy}^{(1)} = 59.8$; trt 2: 137.2, 142.0, 136.0; trt 3: 141.2, 86.8, 103.6. Separate slopes 1.091, 0.991, 0.734.

$$\begin{aligned} SS_E(\mathrm{F}) &= \left(78.8 - \tfrac{59.8^2}{54.8}\right) + \left(142.0 - \tfrac{136.0^2}{137.2}\right) + \left(86.8 - \tfrac{103.6^2}{141.2}\right) \\ &= 13.54 + 7.19 + 10.79 = 31.52 \text{ (9 df)}, \\ F_0 &= \frac{(38.57 - 31.52)/2}{31.52/9} = \frac{3.53}{3.50} = 1.01 < F_{0.05,2,9} = 4.26. \end{aligned}$$

No evidence against parallel slopes: the common-slope ANACOVA is appropriate.

Example 11.7: Fiber strength by machine (Final 2025 setting)

Three machines, $n = 5$, y = strength, x = thickness:

	Machine 1		Machine 2		Machine 3	
	y	x	y	x	y	x
	36	20	40	22	35	21
	41	25	48	28	37	23
	39	24	39	22	42	26
	42	25	45	30	34	21
	49	32	44	28	32	15
Total	207	126	216	130	180	106

$S_{xx} = 261.73$, $S_{xy} = 282.60$, $S_{yy} = 346.40$; $T_{xx} = 66.13$, $T_{xy} = 96.00$, $T_{yy} = 140.40$; $E_{xx} = 195.60$, $E_{xy} = 186.60$, $E_{yy} = 206.00$.

$$\hat{\beta} = 186.6/195.6 = 0.954, \quad SS_E(\mathrm{F}) = 206.0 - \tfrac{186.6^2}{195.6} = 27.99 \text{ (11 df)}, \quad MS_E = 2.544,$$

$$SS_E(\mathrm{R}_\tau) = 346.4 - \tfrac{282.6^2}{261.73} = 41.27, \quad F_0 = \frac{(41.27 - 27.99)/2}{2.544} = 2.61 < 3.98 :$$

no machine difference after adjusting for thickness. Slope: $F_0 = \frac{186.6^2/195.6}{2.544} = 69.97 > 4.84$: thickness is significant. $\hat{\mu} = 40.2$, $\hat{\tau} = (0.182,\ 1.219,\ -1.402)$. Parallelism: $SS_E(\mathrm{F}) = 25.25$ (9 df), $F_0 = \frac{(27.99-25.25)/2}{25.25/9} = 0.49$: parallel slopes accepted.

7 Assumption Checking

Residuals $e_{ij} = y_{ij} - \hat{y}_{ij}$: normal Q–Q plot; residuals vs fitted values; residuals vs treatment; residuals vs x (checks linearity in the covariate). For the cracker data all plots are satisfactory.